

SELECTION SORT:

Selection Sort is a simple comparison-based sorting algorithm. It works by repeatedly selecting the **smallest element** from the unsorted part of the array and placing it at the beginning. This process continues until the entire array is sorted.

Algorithm:

1. Read the number of elements.
2. Enter the array elements.
3. Assume the first unsorted element is the minimum.
4. Compare it with the remaining unsorted elements.
5. Find the smallest element.
6. Swap it with the first unsorted element.
7. Move the sorted boundary one position to the right.
8. Repeat the process until all elements are sorted.
9. Display the sorted array.

Output:

```
Enter the number of elements:
5
Enter the elements:
3
4
1
2
7
Sorted array:
1 2 3 4 7
Process finished with exit code 0
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Time Complexity:

Case	Time Complexity
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Best Case	$O(n^2)$
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Average Case	$O(n^2)$
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Worst Case	$O(n^2)$
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Space Complexity:

Type	Space Complexity
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Selection Sort	$O(1)$
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